THE MAXIMUM NUMBER OF VERTICES OF A $(3,5)$-ZONOBOXTOPE IS 42

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Abstract. A (d,n)-zonoboxtope is a maxout polytope of type $(d,n,1)$: the convex hull of a union of two weighted Minkowski sums of a common family of n segments in $\mathbb{R}^d$. Balakin, Cox, Loho, and Sturmfels (arXiv:2509.21286) prove that a $(3,5)$-zonoboxtope has at most 44 vertices, assert that this bound is tight, and conjecture $4\sum_{k=0}^{2}\binom{n-1}{k}$ as the maximal vertex number of $(3,n)$-zonoboxtopes for all odd n. We prove that the maximum at $(d,n)=(3,5)$ is exactly 42: the bound 44 is not attained, the tightness assertion fails, and the odd-n case of the conjecture fails at its first instance beyond $n=3$. The proof is computer-assisted and fully certified: a symmetry reduction to a single oriented-matroid cell and four split representatives, followed by 132,560 exact Gordan certificates whose multipliers are polynomial in the maximal minors of the configuration, verified modulo the Grassmann–Plücker ideal in exact rational arithmetic. An explicit rational 42-vertex instance is included, together with a stdlib-only verifier, as ancillary files: a two-second command checks the attainment, and four further commands re-verify the entire upper-bound library. In the positive direction we certify that the conjectured values are attained at $(3,8)$ and $(4,6)$, the latter resolving that case completely.

1. Introduction

Maxout polytopes were introduced by Balakin, Cox, Loho, and Sturmfels [1] as the polytopes computed by feedforward neural networks with maxout activation and nonnegative weights after the first layer. The first interesting class beyond zonotopes is the class of polytopes of type $(d,n,1)$, called (d,n)-*zonoboxtopes* in [1, §2]: polytopes of the form

$$Q \;=\; \operatorname{conv}\Big(\sum_{i=1}^{n} a_i I_i \;\cup\; \sum_{i=1}^{n} b_i I_i\Big) \;\subset\; \mathbb{R}^d, \tag{1}$$

where $I_1,\dots,I_n$ are segments and $a,b\in\mathbb{R}^n_{\geq 0}$. Writing $I_i = m_i + [-u_i,u_i]$ with midpoint $m_i\in\mathbb{R}^d$ and direction $u_i\in\mathbb{R}^d$, the parameters are (u_i,m_i,a,b).

For $d=3$, [1, Proposition 6.5] states that a $(3,n)$-zonoboxtope has at most $16,26,44,60$ vertices for $n=3,4,5,6$, and that these bounds are tight. The upper bounds are proven there by a depth-first search over valid bicolorings, performed for each combinatorial type of three-dimensional zonotope; the tightness by extremal examples found by sampling. Part 1 of [1, Conjecture 6.6] proposes, for the maximal vertex number of a $(3,n,1)$-maxout polytope,

$$4\sum_{k=0}^{2}\binom{n-1}{k}\ \text{ if } n \text{ is odd,}\qquad 4\sum_{k=0}^{2}\binom{n-1}{k}-(n-2)\ \text{ if } n \text{ is even,}$$

which equals 44 at $n=5$. Our main result is:

Theorem 1. *The maximum number of vertices of a $(3,5)$-zonoboxtope is 42.*

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Consequently the upper bound 44 of [1, Proposition 6.5] is correct but not attained: its tightness assertion fails at $n = 5$, and the odd-n case of part 1 of [1, Conjecture 6.6] fails at its first instance beyond $n = 3$. In particular no sampled instance can attain 44 at $n = 5$. Our implementation numerically reproduces the other three values $16, 26, 60$ of [1, Proposition 6.5] and certifies exact instances attaining the conjectured values at $(3, 8)$ and $(4, 6)$ (Proposition 3), so we suspect a floating-point vertex-count artifact in the verification of the sampled $n = 5$ example rather than any definitional mismatch; an explicit instance, once equipped with vertex witnesses by any exact convex-hull code, can be checked in seconds in the format of our ancillary verifier. Following [1, Remark 6.7], our upper bound is proven for the wider family of *zonoboxtope candidates* (two zonotopes with correspondingly parallel generators and independent translations — the translation parameter T below is unconstrained), while the 42-vertex witness is a genuine zonoboxtope; so the maximum is 42 for both families. The attainment half of Theorem 1 is the following explicit instance, whose vertex count is verified in exact rational arithmetic by a Python-stdlib program included with this note (Section 3).

Proposition 2. *Let $u_1, \dots, u_5$ be the rows of*

$$U = \begin{pmatrix} -19/26 & 1/30 & -15/22 \\ 0 & -1 & 1/16 \\ 1/19 & 20/23 & -1/2 \\ 10/21 & -6/7 & -6/29 \\ 6/19 & 4/27 & 15/16 \end{pmatrix},$$

let $m_1 = (-1/7,\ 17/28,\ 1/12)$ and $m_2 = \cdots = m_5 = 0$, and let

$$a = \left(\tfrac{47}{25}, \tfrac{49}{20}, \tfrac{9}{4}, \tfrac{14}{17}, \tfrac{23}{18}\right), \qquad b = \left(\tfrac{16}{17}, \tfrac{27}{22}, \tfrac{9}{8}, \tfrac{28}{17}, \tfrac{23}{9}\right).$$

Then the $(3, 5)$-zonoboxtope (1) has exactly 42 vertices.

Verifying this note before reading it. Every claim is machine-checkable, and the reader is invited to check first and read second. The attainment half (Propositions 2 and 3) takes about two seconds with no dependencies beyond the Python standard library: download the ancillary files of this note into one directory and run

```
python verify_c66_new_cases.py        # ~2 s, exact arithmetic -> PASS
```

This proves, in exact rational arithmetic, that the instance above has exactly 42 vertices (and verifies the 110-, 104-, 58- and 84-vertex instances of Proposition 3 and beyond). The upper bound rests on a certificate library whose complete re-verification is four commands from the public repository (`ai/maxout` of the repository cited below; only `check_transport.py` needs anything beyond the standard library):

```
python check_om35_uniqueness.py          # seconds, stdlib
python capstone/check_split_orbits.py    # seconds, stdlib
python capstone/independent_audit.py     # ~2 min, stdlib, all
                                         # 132,560 certificates
python capstone/check_transport.py       # ~2 min
```

Section 3 states precisely what each run proves. The third command is a from-scratch verifier that shares no code and no algebra with the programs that produced the library.

The substance of Theorem 1 is the upper bound. It is a computer-assisted proof in the certificate tradition: the mathematical content is carried by exact, independently checkable certificates — a library of 132,560 Gordan infeasibility certificates valid across an entire oriented-matroid cell, plus small exact computations for the symmetry reduction — while search heuristics (floating-point linear programming, sampling) were used only to *find* certificates, never to justify a claim. Section 2 describes the proof; Section 3 describes how to verify it. The complete certificate library, all generating and checking programs, and a detailed prose account are in the public repository

https://github.com/reuellee/finite-certificates

(directory `ai/maxout`; the capstone document `ai/maxout/capstone/CAPSTONE.md` gives the full argument with file-level pointers).

In the positive direction, the same exact-verification harness confirms two further values asserted or conjectured in [1]:

Proposition 3. *There exist a* $(3,8)$*-zonoboxtope with* 110 *vertices and a* $(4,6)$*-zonoboxtope with* 104 *vertices, both certified in exact arithmetic. In particular the even-n value* 110 *of* [1, Conjecture 6.6] *is attained at* $n = 8$*, and the case* $(4,6)$ *of the conjecture is resolved completely: the maximum there is exactly* 104*, since* 104 *equals twice the maximal vertex number of a* 4*-dimensional zonotope with* 6 *generic generators.*

We also certify 58 vertices at $(4,5)$ and 84 at $(3,7)$ (conjectured maxima: 60 and 88). Extensive searches (complete only for the restricted models they searched, with direction configurations sampled rather than exhausted) found no more; whether the conjectured odd-n values fail for all odd $n > 3$, and by how much, is an open question we do not settle here.

2. The proof

Throughout, $(d,n) = (3,5)$. We outline the chain; every step is either a short argument reproduced here or an exact computation named here and shipped in the repository.

2.1. **Degenerate cases.** Every vertex of a hull of a union (1) is a vertex of one of the two zonotopes, so $f_0(Q) \le f_0(Z^a) + f_0(Z^b)$. The vertex count of a 3-dimensional zonotope equals the chamber count of the central plane arrangement dual to its generators, which is at most $2\big(\binom{4}{0} + \binom{4}{1} + \binom{4}{2}\big) = 22$ for five planes, with equality iff the arrangement is simple; chamber counts of central arrangements are even (chambers come in antipodal pairs), so a non-simple arrangement has at most 20. Hence if the direction configuration $U = (u_1, \dots, u_5)$ is *not* generic (some 3×3 minor vanishes), each copy has at most 20 vertices and $f_0 \le 40$; and if some $a_i = 0$ or $b_i = 0$, that copy has at most $2(1+3+3) = 14$ vertices and $f_0 \le 36$. Both are below 42, so we may assume U generic and $a, b > 0$.

2.2. **The chamber model and the target systems.** For generic U and $a, b > 0$ the two zonotopes are normally equivalent, and (specializing the framework of [1, §§5–6] to two normally equivalent zonotopes) $f_0(Q) \le$ #chambers+#bicolored chambers, where a chamber of the arrangement $\{u_i^\perp\}$ is *bicolored* when it contains linear objective directions maximized on each of the two copies; the inequality is an equality in the strict generic position of §2.5. With 22 chambers this gives $f_0 \le 44$, attainable only with every chamber bicolored.

Write $\delta_i = a_i - b_i$, let $s_i = \operatorname{sign} \delta_i \in \{\pm 1\}$ be the *split* and $w_i = |\delta_i| > 0$ the residual weight; all translation freedom enters through a single unconstrained vector $T \in \mathbb{R}^3$ (for zonoboxtopes (1), an explicit linear combination of the m_i; for zonoboxtope candidates, the relative translation of the two copies). The all-chambers-bicolored condition becomes a strict linear feasibility system in $x = (T, w) \in \mathbb{R}^3 \times \mathbb{R}^5$: for each unordered pair $i < j$, the two antipodal rays $\pm C_{ij}$, $C_{ij} = u_i \times u_j$, each carry a sign $\sigma_{ij,\pm} \in \{\pm 1\}$, and the system is

$$\sigma_{ij,\pm}\Big(\pm\langle C_{ij}, T\rangle + \sum_{t\notin\{i,j\}} s_t\, D_{tij}\, w_t\Big) > 0 \quad \text{(20 rows)}, \qquad w_t > 0 \quad \text{(5 rows)}, \tag{2}$$

where $D_{tij} = |\det(u_t, u_i, u_j)|$. Not every sign pattern $\sigma \in \{\pm 1\}^{20}$ is consistent with all 22 chambers seeing both signs: the chamber/ray incidence, a combinatorial invariant of the chirotope of U, admits exactly 33,140 *valid* patterns (three independent enumerations agree). A $(3, 5)$-zonoboxtope with 44 vertices and strict side signs therefore yields a valid σ, a split s, and a strictly feasible system (2).

2.3. **Cell-wide Gordan certificates.** A vector $y \geq 0$, $y \neq 0$ with $B^{\mathsf{T}} y = 0$, where B is the 25×8 coefficient matrix of (2), proves strict infeasibility (Gordan's theorem). We construct such certificates *cell-wide*: the 25 multipliers are polynomials in the ten quantities D_{tij} — side multipliers of degree ≤ 2, weight multipliers of degree ≤ 3, all coefficients nonnegative, at least one positive — such that $B^{\mathsf{T}} y = 0$ holds *identically modulo the Grassmann–Plücker ideal* given the chirotope's signs. Such a certificate specializes to a valid Gordan vector at *every* configuration realizing the chirotope: on a generic configuration every $D_{tij} > 0$, so nonnegative coefficients give $y \geq 0$ and the positive coefficient gives $y \neq 0$.

Two mechanisms produce the library. If some pair class (i, j) has equal ray signs σ^* with $\sigma^* s_t = -1$ for all $t \notin \{i, j\}$, the closed form $y_{ij,\pm} = 1$, $y_{w,t} = 2D_{tij}$ is a certificate with no Grassmann–Plücker reduction needed (and, at the prefix splits $k \in \{1, 2\}$, this *single-class family* is provably the complete coefficientwise-positive mechanism among ordinary polynomial identities, i.e. those requiring no Grassmann–Plücker reduction). All remaining systems are killed by explicit degree-$(2, 3)$ certificates found by exact kernel computations in the quotient ring — normal forms modulo a Gröbner basis of the signed-determinant Plücker ideal over $\mathbb{Q}$ — with typical supports of four terms. Deliberately impossible control systems ("canaries"), a discipline instituted after one such control exposed a verifier defect in an earlier stage, were wired into the $k \in \{1, 2\}$ completion sweep and the $\{1, 3\}$-split sweep and correctly rejected throughout; the $k = 0$ sweep ran without canaries, and its certificates are covered by the independent audit of Section 3 and were re-verified at fresh realizations in review.

2.4. **Symmetry: one cell, four splits.** The group $G = S_5 \ltimes \{\pm 1\}^5$ acts on configurations by $u_i \mapsto \varepsilon_i u_{\pi^{-1}(i)}$. The action preserves the polytope: reorientation fixes each segment $m_i + w_i[-u_i, u_i]$ as a set, and relabeling renames segments (carrying m, a, b, s along). Two exact computations reduce the quantifiers:

(i) The uniform rank-3 chirotopes on 5 elements number exactly 384 and form a *single* G-orbit. (Only the containment direction is used: the chirotope of any real generic configuration satisfies the three-term sign conditions, hence lies among the 384. That all of them are in fact realizable [2, Ch. 8] is reassurance, not a step of the proof.) Hence every generic instance moves, without changing its polytope, to an instance whose configuration realizes one fixed reference chirotope χ_{ref} — that of the integer configuration U_{ref} with rows $(-6, -13, 18)$, $(-9, -12, 8)$, $(-13, -4, 16)$, $(4, -19, -8)$, $(16, 15, -12)$.

(ii) The stabilizer of χ_{ref} in G has order 10 and faithful permutation image the dihedral group D_5 (the automorphism group of the pentagon). Its orbits on split subsets $A = \{t : s_t = +1\}$ are: $\{\emptyset\}$; all five singletons; *two* orbits of 2-subsets, represented by $\{1, 2\}$ and $\{1, 3\}$; and complements. Swapping the two copies reduces complements via an exact row correspondence (the *flip*). Hence every instance normalizes to χ_{ref} with split subset $\emptyset$, $\{1\}$, $\{1,2\}$, or $\{1,3\}$.

We emphasize (ii): treating splits "by size" is *not* sound — the two orbits of 2-subsets are genuinely inequivalent — and an earlier version of our own library covered only one of them. The gap was found while writing out the transport argument and closed by a fourth sweep; we flag it because size-based split reductions look innocuous and are wrong.

The certificate library consists of an exact cell-wide certificate for each of the 33,140 valid patterns at each of the four split representatives: 132,560 certificates in total, of which 74,923 are closed-form single-class and 57,637 are explicit quotient-ring certificates. Certificates transport along G covariantly: the ray pair of class (i,j) maps to that of class $(i',j') = (\pi(i),\pi(j))$ twisted by the sign $\tau = \pm\varepsilon_{i'}\varepsilon_{j'}$ (negative iff π reverses the order of the pair), the D-variables permute, and valid patterns map bijectively to valid patterns. The bookkeeping is verified exactly on thousands of transported certificates against independently built systems, and the validity enumeration is verified to be preserved by the full stabilizer.

2.5. **Genericity, parity, and the endgame.** Two boundary issues remain: the normal form requires all $\delta_i \neq 0$, and the chamber-model count requires no accidental coincidence among the 2^6 candidate points. Both are handled by a perturbation lemma, stated in *candidate coordinates* — two independent translations $c_A, c_B \in \mathbb{R}^3$ and two width vectors $a, b \in \mathbb{R}^5_{>0}$, which subsume the shared-midpoint parameterization of (1): every vertex of Q is strictly exposed among the candidate points, and candidates vary polynomially in (c_A, c_B, a, b), so all vertices persist under small parameter perturbations; the excluded loci (a vanishing width difference $a_i - b_i$, a vanishing side value, a coincidence of candidates that are not identical as polynomial maps) are finitely many proper algebraic subsets, whose union misses arbitrarily nearby parameters. Hence any candidate instance with $f_0 \geq 43$ yields a nearby *strict generic position* instance with $f_0 \geq 43$.

In strict generic position the count $f_0 = 22 + \#\text{bicolored}$ is literal, and the bicolored count is even: the zero set of the support-function difference crosses chamber walls transversally, meets each open chamber cone in a single connected sector (the intersection of a plane with a convex cone is convex), and traces cycles in the chamber-adjacency graph, which is bipartite (the chambers of a central simple arrangement of five planes 2-color by sign-vector parity). So $f_0 \geq 43$ forces $f_0 = 44$ with all side signs strict — producing a valid pattern and a strictly feasible system (2) at some realization of χ_{ref} with one of the four split representatives, which the certificate library kills. Hence $f_0 \leq 42$ for every zonoboxtope candidate.

Proof of Theorem 1. The upper bound is the chain of §§2.1–2.5; the attainment is Proposition 2. □

3. Verification

Ancillary files. The ancillary files of this note contain the five instance certificates (`cert_35_42.json`, `cert_38_110.json`, `cert_46_104.json`, `cert_45_58.json`, `cert_37_84.json`) and the stdlib-only verifier `verify_c66_new_cases.py` ($\sim$2 seconds), which proves in exact `Fraction` arithmetic that each named instance has *exactly* the claimed vertex count: all 2^{n+1} candidate sign points are pairwise distinct, each claimed vertex carries a strict witness direction, and each non-vertex carries an explicit convex combination. This suffices to verify Propositions 2 and 3 independently of everything else.

The upper-bound library. From the repository, directory `ai/maxout`: `check_om35_uniqueness.py` (the 384-chirotope single-orbit computation; stdlib), `capstone/check_split_orbits.py` (the stabilizer and split-orbit accounting; stdlib), `capstone/check_transport.py` (exact transport verification), and `capstone/independent_audit.py`, which re-proves *every* certificate in the library.

That last program is written from scratch against the serialized artifacts: it imports nothing from any of the programs that produced them and uses only the Python standard library. It does not re-use the generators' quotient-ring construction — there is no Gröbner basis and no normal form anywhere in it. Instead it decides cell-wideness by the observation that the configurations realizing a fixed chirotope form a nonempty *open* subset of $\mathbb{R}^{15}$, so an identity in the D_{tij} holds across the cell if and only if its pullback under $D_T \mapsto \chi_T \det_T(x)$ is the zero polynomial in the 15 entries of a symbolic 5×3 configuration x — an equivalence, whereas membership in the Plücker ideal is a priori only sufficient. That vanishing is decided by exact integer evaluation against a family of evaluation functionals whose sufficiency the run certifies itself: writing M_e for the number of degree-e monomials in ten variables, E for the evaluation matrix and J for the span of the monomial multiples of the five signed three-term Plücker relations, one always has $\operatorname{rank} E + \operatorname{rank} J \leq M_e$, and the run computes both ranks and finds equality, which forces $\ker E$ to be exactly the degree-e part of the kernel of the pullback. Nothing about Hilbert functions or standard monomials is assumed; the classical values $10, 50, 175, 490$ for $e = 1, \ldots, 4$ come out as confirmed predictions.

The same run re-derives the reference chirotope, the 22 chambers (with a completeness proof) and the 33,140 valid side patterns from the reference matrix alone; verifies for every certificate that the coefficients are nonnegative and not all zero, that all ten identities hold across the cell, and that the specialization is an exact Gordan vector at U_{ref} and at two fresh integer realizations of the chirotope; verifies that the library *covers* every valid side pattern at each of the four splits, with no gap and no duplicate; and runs a canary battery in which ten deliberate corruptions are all rejected, recording which layer caught each. Two of the canaries test precisely what a generator-coupled auditor cannot: a mutation that is a valid Gordan vector at U_{ref} but not cell-wide, and a systematic falsification of the symbolic semantics (the whole signed-determinant construction rebuilt with the wrong chirotope, certificates untouched). It reports `132869 checks OK, 0 failed` in about two minutes, and writes an immutable JSON report with the counts, the per-degree rank certification, the coverage differences, the canary table and the SHA-256 of every input artifact.

A second, older auditor, `stage2c2_gpt/audit_all_certificates.py`, checks the same library (132,681 checks including redundant coverage; zero failures; about 20 minutes, and it needs `numpy/scipy /sympy`). It is retained and agrees with the independent audit on every shared count, but it is *not* independent of the generators: it imports their quotient-ring construction, and its own independently written half checks only specialization at U_{ref}. The generation-side programs, sweep artifacts, and a complete prose account with erratum history are in the same directory.

Provenance. This result was produced in a multi-model AI research pipeline: search, certificate generation, and drafts of the arguments were carried out by large language models (GPT-5.6, Gemini 3.1 Pro, Claude) directing exact computer algebra, with adversarial cross-review between models, canary controls in the certificate sweeps, and independent re-verification of every serialized object. No mathematical claim in this note rests on *unverified* model output: the model-produced certificates are exactly re-checked by the independent audit described above — which was written specifically so that no program on the generation side is trusted — the upper bound rests on that audited library and the short arguments of Section 2, and the attainment on the ancillary stdlib verification. What the audit necessarily takes as given is the 25×8 row model (2) itself, since that is the statement of what the certificates certify; it is short enough to check by reading. The author directed the program and takes responsibility for the claims.

Acknowledgments. I thank the authors of [1] for a beautiful paper. Any explicit 44-vertex $(3,5)$-instance would contradict Theorem 1, and, once equipped with vertex witnesses by any exact convex-hull code, could be checked in seconds in the format of the ancillary verifier; I would be grateful for any such communication.

References

[1] A. Balakin, S. Cox, G. Loho, B. Sturmfels, *Maxout polytopes*, preprint, 2025, arXiv:2509.21286.
[2] A. Björner, M. Las Vergnas, B. Sturmfels, N. White, G. M. Ziegler, *Oriented Matroids*, 2nd ed., Encyclopedia of Mathematics and its Applications 46, Cambridge University Press, 1999.

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